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The line $2x + 3y = 12$ intersects the curve $y^2 = 4x - 8$ at points A and B .

Find the value of p and q for which the length of AB can be expressed as $p\sqrt{q}$.

[6]

From ①, $2x = 12 - 3y$ ——— ③

Subst. ③ into ②,

$$y^2 = 2(12 - 3y) - 8 \quad (M1)$$

$$y^2 = 24 - 6y - 8$$

$$y^2 + 6y - 16 = 0$$

$$(y + 8)(y - 2) = 0 \quad (M1)$$

$$y = -8 \quad \text{or} \quad y = 2$$

$$2x = 12 - 3(-8)$$

$$2x = 36$$

$$\therefore x = 18$$

$$\therefore A(18, -8) \quad (A1)$$

$$2x = 12 - 3(2)$$

$$2x = 6$$

$$x = 3$$

$$B(3, 2) \quad (A1)$$

$$\text{Length of } AB = \sqrt{(18-3)^2 + (-8-2)^2}$$

$$= \sqrt{225 + 100}$$

$$= \sqrt{325}$$

$$= \sqrt{25 \times 13}$$

$$= 5\sqrt{13} \text{ units}$$

$$= p\sqrt{q}$$

$$\text{By comparison, } p = 5, q = 13. \quad (A2)$$